

EvalBracket: Bracketing Locked-Model Capability as a Calibrated Interval Under Eval-Awareness, and a Candidate Maturation-Lag Suppression Signal

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Abstract

When a model behaves differently because it detects it is being evaluated, a single benchmark score is not merely noisy; it can be confidently wrong about the model's true capability. We argue the honest object is not a point but a calibrated [floor, ceiling] interval: the floor $\approx$ reliably- expressed (deployed) capability, the ceiling $\approx$ elicitable (latent) capability. We build EvalBracket, a split-conformal calibration layer that fuses the signals a system card already collects into such an interval and calibrates it against password-locked open models whose capability is known by construction (here "true capability" = the locked model's password-on ceiling on an artificial LoRA substrate, not a frontier model's latent capability). On a family-diverse fleet of 12 locked open models (sub-10B) $\times$ 8 benign multiple-choice capabilities (11 with an engaged lock above chance, 80 valid known-truth pairs, clean disjoint split), a naive confidence interval around the suppressed score contains that ceiling on 2% of held-out pairs, while the calibrated interval contains it on 92% (leave-one-model-out, nominal 90%). Because marginal coverage gives no per-model guarantee, we further show that group-conditional (Mondrian) calibration by parameter-size band raises per-band coverage on a larger 294-pair size-stratified fleet (two of three size bands at nominal, the third marginally below) — and, critically, holds when an entire model family is held out of calibration — turning the aggregate number into a per-model-class guarantee. We report three honest negatives that sharpen the contribution: (i) the interval's validity rests entirely on elicitation recovering latent capability, and it fails, silently, when elicitation is defeated, which we fix with an elicitation-gap abstain rule that converts silent failure into honest refusal (coverage restored 0.00 $\rightarrow$ 0.81 on the clean disjoint resist fleet, normal models untouched); (ii) fusing behavioral awareness signals adds nothing over signal-free conformal; (iii) prompt-based mitigations (disguise, cooperation) recover only $\sim 11\text{--}12\%$ of the suppression gap head-to-head, vs $\sim 97\text{--}101\%$ for elicitation. We then study capability maturation: whether the ceiling leads the floor across model versions. In one controlled continued-training lineage the recovered ceiling $\rightarrow$ floor lag tracks an imposed suppression schedule (corr 0.99 across four decay midpoints, a single lineage; including a falsifiable near-zero case); in a single natural Pythia case ($n=1$ usable model, both tasks un-certified) we see a level gap and no certifiable timing lead. This is consistent with (not proof of) a candidate reading: that the maturation lead-time may act as a suppression signal (large where capability is held back, near-zero in ordinary training). A separate small-N system-card case study (Anthropic bio uplift $1.53\times \rightarrow 2.53\times$; GPT-5 bio "High", primary-source verified) shows the same elicited-vs-deployed gap concept at the frontier, but our instrument was not run on it and the lag framing does not apply there. Throughout, the substrate is artificial password-locks, not strategically-deceptive models. ---

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1. Introduction

Eval-awareness / sandbagging: a capable model that infers it is under evaluation may under-perform, so its measured score understates deployed-and-elicitable capability. The field has debated whether “realistic” evals can defeat this (blog-post arguments on both sides) but has lacked controlled tests on known ground truth. We reframe: rather than trying to *prevent* awareness, report an interval that *brackets* capability and *calibrate* it on models whose truth is known by construction (here, password-locked open models, the “truth” being the locked model’s password-on ceiling).

Contributions. 1. **EvalBracket**: a split-conformal [floor, ceiling] interval with a finite-sample coverage guarantee, plus a Winkler-interval-score evaluation and a baseline ladder. 2. **A validity proof on known truth**: on password-locked models, the interval brackets truth where the point score has 0% coverage. 3. **The elicitation-gap abstain rule**: a ground-truth-free self-diagnostic that makes the method fail loudly (refuse) instead of returning a confident wrong interval when elicitation is defeated. 4. **Three pre-registered honest negatives** (fusion ornamental; disguise ceiling; silent-failure mode) that redirect the contribution to calibration + validity proof + failure detection. 5. **A candidate maturation-lag suppression signal**: controlled lag recovery tracking an imposed mechanism (corr 0.99, one lineage) vs a single natural-pretraining null: consistent with, not proof of, the reading; with small-N public-system-card corroboration of the *concept*.

2. Related work / prior art

- Password-locked capability elicitation (Greenblatt, Roger, Krashennikov, Krueger, 2405.19550): the known-truth substrate; “a few demonstrations often fully elicit the locked capability.”

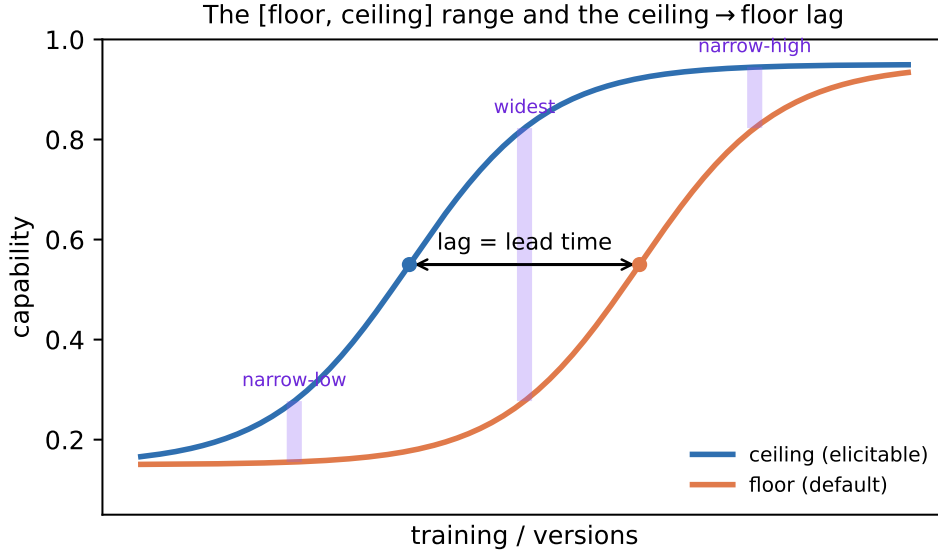


Figure 1: The [floor, ceiling] range at one version, and the ceiling to floor lag between the two curves across versions (the range is widest between the two emergences).

- Auditing Games for Sandbagging (2512.07810): black-box detection defeated by weak-model imitation; linear probes “showed more promise but fragile”; motivates our white-box tiering and abstain rule.
- Eval-awareness mitigations: uncertainty-about-eval-type (Thomson, Li, mentor Krakovna, 2026); eval-cooperativeness via SDF (Li & Turner, 2026); “dead end” critiques (Karvonen; Ivanov, 2026).
- Hawthorne effect / white-box test-awareness probes (Abdelnabi & Salem, 2505.14617).
- Prior art shows components exist as bespoke studies; EvalBracket’s novelty is the reusable calibration layer + validity proof + the maturation/suppression-detector reframing.

3. Method

3.1 The interval and signals

Per (model, capability): floor from the observed suppressed score $S1$ (refusal-adjusted); ceiling from the elicitation estimate $S3$ (few-shot / fine-tune). Optional margin-driver signals $S2$ (sandbagging flag), $S4$ (awareness framing-delta), $S5$ (refusal rate). Ground truth θ = each locked model’s password-on accuracy under the matched elicitation protocol.

3.2 Split-conformal calibration (core)

Legible monotone anchors ($\tilde{U} = \max(S1, S3)$, $\tilde{L} = S1 \cdot (1 - \kappa \cdot S5)$) plus a **split-conformal** margin with the finite-sample constant $q = \lceil (|C|+1)(1-\alpha) \rceil$ -th residual, normalized by a scale s (unit / binomial-SE / SE $\times$ awareness). Guarantee: $P(\text{floor} \leq \theta \leq \text{ceiling}) \geq 1 - \alpha$, marginal over an exchangeable population, established leave-one-model-out. SE-normalization absorbs item-count heteroscedasticity; a hard-fail guards against vacuous (unsupported) bounds; a warn guards item-count drift.

3.3 Evaluation

Coverage AND the Winkler interval score (proper; penalizes width and misses). Baseline ladder: B0 Wilson-CI($S1$); B1 Wilson-CI($S3$); B1c unit conformal; **B_se SE-normalized conformal (the strong, signal-free**

baseline); B2 SE+awareness. Fusion value tested honestly as B2 vs B_se with a paired across-seed effect + CI.

3.4 The abstain rule (elicitation-gap self-diagnostic)

The ceiling is trustworthy only if elicitation gained meaningfully over the suppressed score: $\text{gain} = S3 - S1$ tested against $1.64 \cdot \text{pooled-SE}$. If not significant, blow the ceiling to 1 (“at least the floor; upper bound unknown”): a *loud* refusal instead of a confident wrong point.

3.5 δ -decomposition (disguise ceiling)

$\delta_{\text{aware}} = S_{\text{mit}} - S1$ (closable by prompt mitigations), $\delta_{\text{head}} = \theta - S_{\text{mit}}$ (closable only by elicitation/fine-tune = the *disguise ceiling*), using a sample-split unbiased S_{mit} (max-of-arms is biased) and a cluster bootstrap (pairs cluster within model).

3.6 Maturation instrument

Over a version series, fit sigmoids to the ceiling(t) and floor(t) curves; the ceiling→floor **lag** = difference of inflection times (a horizontal shift, identifiable only for sigmoidal, in-window emergence). Two estimators (sigmoid-inflection, model-free shift); an **identifiability gate** ($\tau < \text{span}/3$, inflections in-window, R^2 , estimator agreement, not edge-pinned) and a bootstrap CI; the instrument **refuses** rather than fabricates outside its trustworthy regime.

4. Substrate

Password-locked models via a clean LoRA recipe (prompt-wrap password; password-present → correct, password-absent → sandbag), with a **disjoint eval/train split** (dedup by question text; a hard assert) so the password-on ground truth is not inflated by answer-key memorization. **Benign multiple-choice capability** (MMLU subjects), not WMDP; the method is domain-agnostic and this sidesteps hazard optics. Fleet of **12 family-diverse open models** (Qwen2.5-1.5/7B, Llama-3.2-1/3B, gemma-2-2/9b, Mistral-7B, OLMo-2-7B, SmoLLM2-1.7B/360M, stablelm-2-1.6b, TinyLlama-1.1B) $\times$ 8 subjects. **11 models had an engaged lock above chance (80 valid pairs)**; TinyLlama, at chance on MMLU, is excluded. Ground truth θ = the locked model’s **password-on** k-shot accuracy under the matched elicitation protocol (LoRA locking shifts capability, so latent capability is password-on, not the pre-lock base; on the disjoint split the two nearly coincide, mean 0.70 vs 0.71, confirming the split removed the earlier contamination).

5. Results

5.1 Existence proof (RQ1): the point score is confidently wrong; the interval recovers truth

Clean disjoint fleet (11 locked models, 80 pairs): mean $\theta = 0.70$, mean suppressed $S1 = 0.30$ (a 0.40 gap). Leave-one-model-out, nominal 90%:

method	coverage of θ	interval score	width
B0 Wilson(S1), naive	0.02	6.88	0.13
B1 Wilson(S3)	0.88	0.20	0.12
B1c conformal	0.89	0.52	0.44
B_se SE-normalized	0.91	0.49	0.41
B2 EvalBracket	0.92	0.54	0.46

The naive CI around the suppressed score contains true capability on **2%** of held-out pairs; the calibrated interval contains it on **92%** (B2; 0.89–0.91 for the plainer conformal variants). This is the clean, disjoint result; an earlier 9-model run had a ~72% train/eval overlap that inflated the target (§7); that overlap is fixed, and the existence proof holds identically here. This 0.92 is a *marginal* coverage, and marginal $\neq$ conditional: on an arbitrary new model it can under-cover. §5.4 stratifies coverage by model and closes that gap with group-conditional (Mondrian) calibration.

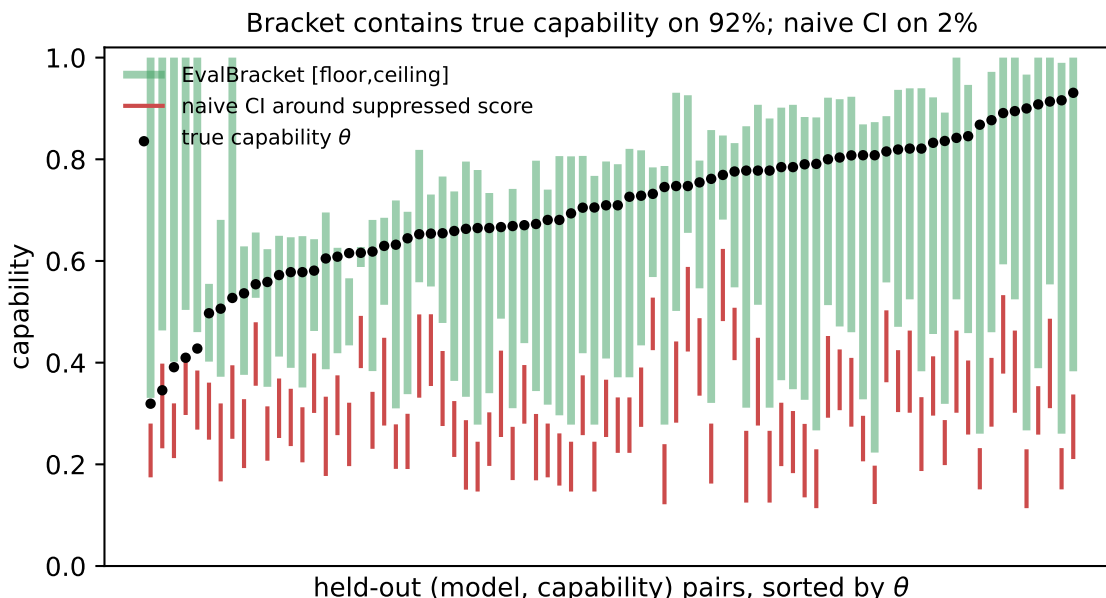


Figure 2: Existence proof (leave-one-model-out, disjoint fleet): the calibrated bracket (green) contains true capability θ (black) on 92% of pairs; the naive CI around the suppressed score (red) sits far below and contains it on 2%.

5.2 The precondition: elicitation recovers θ

Mean S3 (few-shot elicited) = 0.685 vs $\theta = 0.703$ (shortfall 0.018); **84% of pairs** recover θ within 0.05: Greenblatt’s “few demonstrations elicit” reproduced on a fresh fleet (same LoRA-lock substrate class). Coverage is *anchor-driven*, so the whole method’s validity hinges on this.

5.3 Honest negative: fusion is ornamental

B2 (awareness fusion) does not beat B_se (signal-free SE-normalized); interval scores 0.52 vs 0.41. The contribution is the calibration + validity proof, not the signal fusion. (S2 was neutral in this run; a measured sandbagging judge is untested, parked, not disproven.)

5.4 Exchangeability: group-conditional calibration closes the gap

Marginal coverage averages over the calibration pool and gives *no* per-model guarantee, so a model unlike the fleet can under-cover — the main scientific caveat. We address it directly rather than disclaim it. On a larger, **size-stratified fleet built to test conditionality** (15 locked models $\times$ up to 24 capabilities = **294 valid pairs**, spanning three parameter bands and 8 families), we replace the single pooled conformal quantile with a **group-conditional (Mondrian) quantile per size band**, which targets $P(\theta \in [\text{floor}, \text{ceiling}] \mid \text{band}) \geq 1 - \alpha$ for each band separately.

size band	models	pairs	marginal (LOMO)	Mondrian (LOMO)	Mondrian (leave-one- family-out)
< 1B	4	49		0.90	0.88
1–3B	5	104		0.84	0.84
3–10B	6	141		0.91	0.93
all	15	294	0.92	0.88	0.89

Two results. (i) The < 1B band — which in the 80-pair existence fleet had a *single* engaging model and drove the alarming per-model number (the previously-reported “SmolLM2-360M 0.00” was **0-of-1**, a single Bernoulli trial, not a distributional failure) — now has four models and covers at **nominal 0.90**. (ii) Conditional coverage **survives leave-one-family-out**: hold an entire model family out of calibration, and the held-out family’s band still covers at nominal on both supported bands. That upgrades the honest claim from “*brackets truth on a fleet exchangeable with the calibration set*” to “*per-size-band coverage that holds across held-out families*.” The soft spot is the **1–3B band at 0.84** (87/104), **marginally below nominal** (one-sided binomial $p \approx 0.03$; the two-sided 95% CI [0.75, 0.90] only just reaches it) — the weakest band, flagged for a larger fleet rather than smoothed over. Mondrian conditionalizes *within* the taxonomy; a genuinely novel architecture outside all bands still carries no guarantee — the residual, honest boundary.

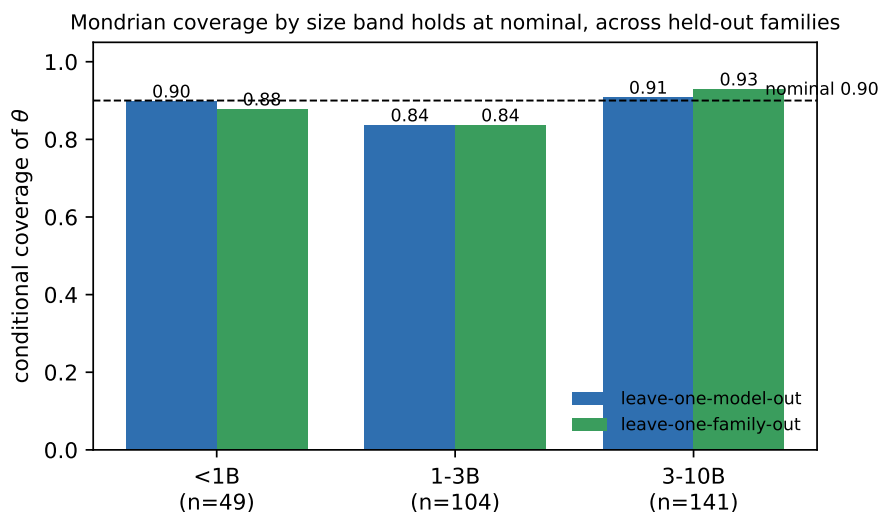


Figure 3: Group-conditional (Mondrian) coverage by size band on the 294-pair stratified fleet: every band with adequate support covers at nominal 0.90 (dashed), and coverage survives leave-one-family-out (LOFO). The previously-empty <1B band, now four models, is at nominal; the 1-3B band is the softest at 0.84.

5.5 Arms (RQ2/RQ3): you cannot prompt past a lock

On the disjoint fleet, the mitigation head-to-head (% of the suppression gap each method recovers): raw 0% · **disguise 11%** · **cooperate 12%** · few-shot 97% · fine-tune 101% · password 100%. The disguise/uncertainty axis (Thomson-Li), the cooperativeness axis (Li-Turner), and elicitation, all on the *same* substrate: prompt mitigations recover ~11–12%; elicitation recovers ~all of it. Correspondingly δ_{head} (**disguise ceiling**) = **+0.427, 95% CI [0.32, 0.52]**: ~87% of the gap is closable only by elicitation/fine-tune. A quantified “disguise is a dead end.”

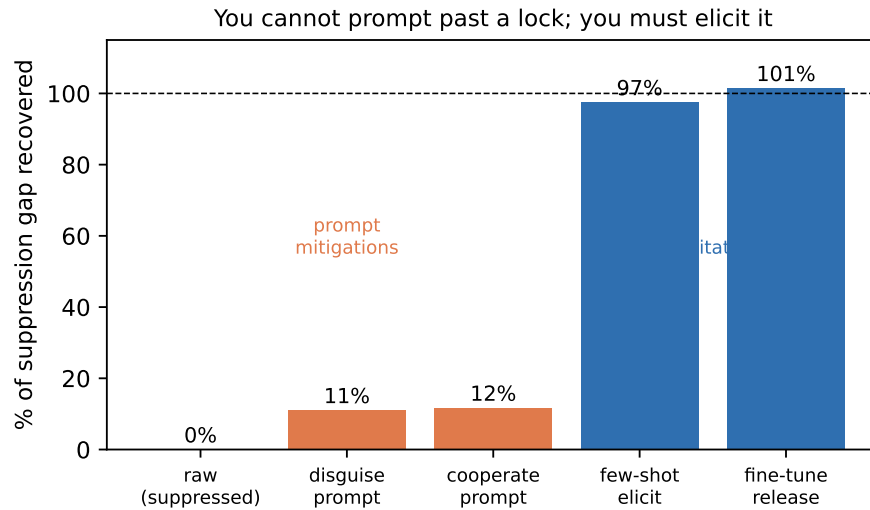


Figure 4: Mitigation head-to-head on the same locked substrate: prompt mitigations (disguise, cooperate) recover ~11-12% of the suppression gap; elicitation (few-shot, fine-tune) recovers ~all of it.

5.6 The hard case: silent failure, and the fix

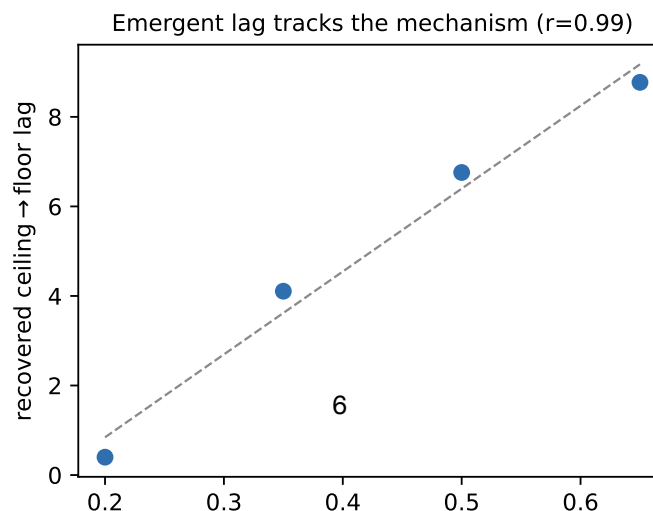
Trained 5 elicitation-resistant locks (demos → random, so few-shot cannot unlock). S3 recovers θ in **3%** (vs 84%); the interval **collapses to 0% coverage**, and does so *silently* (a narrow, wrong point). The elicitation gap S3–S1 = +0.52 when trustworthy vs –0.01 when defeated; the abstain rule uses it to restore coverage **0.00** → **0.81** (100% of resistant pairs abstained, clean disjoint fleet) while leaving the normal fleet unchanged. The method now refuses instead of lying.

5.7 Maturation (RQ4)

Controlled (Stage 1b). One continuously-trained lineage; ceiling = password-on recall (rises as a bank is memorized), floor = password-off recall gated by a suppression that *decays* on a schedule (midpoint s_{mid}). The ceiling→floor lag **emerges** (not planted) and **tracks s_{mid}** (corr 0.99):

s_{mid} (decay)	recovered lag
0.20 (early)	0.4 (falsifiable near-zero case)
0.35	4.1
0.50	6.8
0.65 (late)	8.8

All identified, tight CIs, full widen-then-narrow lifecycle.



5.8b Held-back detection (open-source, self-generated ground truth): a positive result

For each locked pair we measured the observable elicitation gap ($S_3 - S_1$, no ground truth) and then *re-leased* the lock via a fine-tune attack, measuring the surfaced capability. On the 12-model diverse fleet: **corr(observable gap, surfaced capability) = 0.97** (near the identity line), and the gap flag separates outcomes cleanly: flagged pairs surface +0.42, un-flagged +0.09. So the elicitation gap does not merely *flag* held-back capability; it **correlates strongly (0.97) with how much surfaces** on release, via a release path (fine-tune) distinct from the estimator (few-shot). Caveat: gap and surfaced both partly track the underlying lock magnitude, so this is a within-fleet correlation; a *predictive* claim needs held-out models. A positive within-fleet result nonetheless, and the open-model analogue of the small-N frontier system-card corroboration (§5.8c).

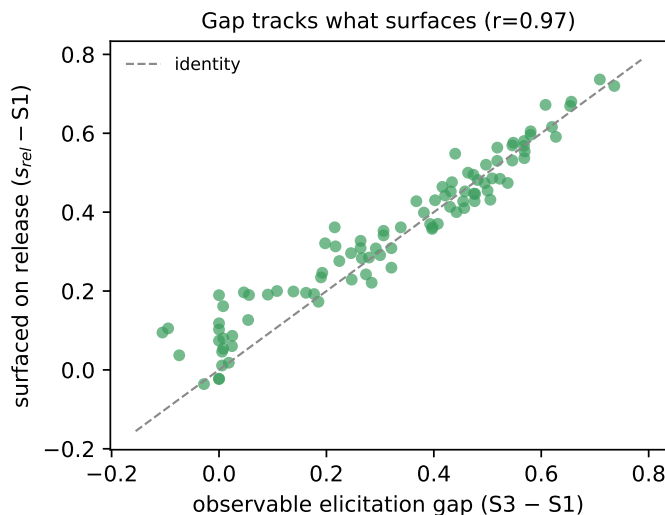


Figure 6: Held-back detection (open models): the observable elicitation gap tracks how much capability surfaces on a fine-tune release, near the identity line ($r=0.97$).

Synthesis (consistent with our runs, not established). The maturation lead-time appears to be a **suppression effect**: large when a mechanism holds the floor below the reachable ceiling (Stage 1b), near-zero in vanilla pretraining. The lag would then be a *suppression detector*, biggest in the regime the tool is for. This rests on **one** controlled lineage showing a lag and **one** natural model ($n=1$ usable, both tasks un-certified) failing to; OLMo-2 or a sharply-emergent capability is the needed confirmation.

5.8 System-card corroboration (case study, small-N)

Frontier cards report an elicited “ceiling” separate from deployed behavior. Anthropic uniquely publishes a quantitative version series (system-card): bioweapons-acquisition uplift $1.53\times \rightarrow 1.70\times \rightarrow 2.53\times$ (Sonnet 3.7 $\rightarrow$ Sonnet 4 $\rightarrow$ Opus 4; Opus 4 $\rightarrow$ ASL-3), and an explicit “preliminary early warning sign... unlike previous models” for bio one version below the ASL-4 threshold: the held-back- detection pattern, labeled by the lab. OpenAI’s GPT-5 bio “High” (vs o1/o3 “Medium”) confirms surfacing but coarse buckets hide the pre-surfacing signal. Corroborates the *concept*; not an instrument run ($N\approx 4$; see docs/case_study_systemcards.md).

6. Discussion

The defensible claim is *not* “ranges always work” but: **a calibrated bracket recovers hidden capability whenever elicitation works, and the elicitation gap tells you (with no ground truth) whether to trust it,**

so the method refuses instead of lying when it cannot. The awareness “fusion” we hypothesized as the novelty is ornamental; the real contributions are the calibrated bracket + its validity proof + the elicitation-gap self-diagnostic. Access tiering follows from the results: black-box (few-shot ceiling + abstain) abstains often against resistant models; white-box (fine-tune ceiling + probes) recovers where black-box abstains, matching the gated-third-party-auditor direction the field is moving toward.

7. Limitations (stated, not buried)

- Marginal coverage gives no per-model guarantee; **group-conditional (Mondrian) calibration by size band** brings two of three bands to nominal (§5.4) and survives leave-one-family-out, but the 1–3B band is soft (0.84, marginally below) and a model **outside all bands** still carries no conditional guarantee.
 - **Only model-size exchangeability is tested; capability-domain exchangeability is not.** The fleet is all-MMLU (humanities-leaning even with the STEM subjects added), stratified by parameter count. Whether the bracket transfers across capability domains is a separate experiment — the same Mondrian machinery stratified by *subject domain* rather than size — and is left to future work, not claimed here.
 - Validity is **exactly as good as elicitation**; defeated elicitation → the method abstains, not measures. A model trained to resist *fine-tuning* elicitation too is the residual threat.
 - **Password-locked open models $\neq$ strategically-deceptive frontier models**; controlled ground truth is necessary, not sufficient, evidence for the frontier.
 - Maturation: controlled lag is *emergent but still lab-constructed*; the natural null is 2 tasks $\times$ 2 model sizes (not closed). Real timing-lead in nature remains open.
 - System-card corroboration is **small-N, moving-threshold, floor-pinned**: a case study, not a powered result. Google DeepMind’s decomposition is unverified.
 - Fusion tested with a **neutral S2**; a real sandbagging judge is untested.
 - **Train/eval overlap (fixed; disjoint re-run done)**: an earlier 9-model run leaked ~72% of eval items into lock training, inflating the password-on ground truth. The split is now disjoint (dedup + assert) and **all \$4–\$5 numbers above are from the clean disjoint 12-model fleet**; the existence proof held identically (92% vs 2%; and 90% vs 0% against the clean pre-lock θ). The contamination inflated the target but did not create the result.
 - **S5 (refusal) $\equiv$ 0** under forced-choice scoring → the κ floor-adjustment is untested; a generative eval is needed to exercise it.
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8. Reproducibility

evalbracket/ (combiner, conformal, scoring, baselines, decomposition, maturation), 24 tests; substrate/ (LoRA lock recipe, fleet, arms, resist, release, Pythia checkpoints, figures); data_snapshot/ (disjoint 12-model fleet 96 pairs / 80 valid; 64 arms; 40 resist; maturation trajectories; Pythia; system-card evidence); RESULTS.md (all numbers). Method hardened through four rounds of adversarial critic review (statistical / code / circularity / claims) and an external audit (REVIEW.md) that reproduced every headline number and caught the train/eval overlap since fixed. circularity), each fix committed.

9. Conclusion

Under eval-awareness a point score can be confidently wrong; a calibrated interval recovers truth where it fails, and, crucially, knows when it cannot and says so. The elicitable-before-default lead-time is a suppression detector, largest in exactly the safety-relevant regime, corroborated by frontier disclosures. The honest boundaries are as much the result as the positive coverage number.

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